

FCFS CPU Scheduling

Aim

To write and execute a Python program to implement the First-Come, First-Served (FCFS) CPU scheduling algorithm and calculate the waiting time and turnaround time for each process, along with their averages.

Program

```
GNU nano 8.7.1                                fcfs.py *
def find_wt(process, n, bt, at):
    wt = [0] * n
    service_time = [0] * n
    service_time[0] = at[0]

    for i in range(1, n):
        service_time[i] = service_time[i - 1] + bt[i - 1]
        wt[i] = service_time[i] - at[i]

        if wt[i] < 0:
            wt[i] = 0
            service_time[i] = at[i]

    return wt

def find_TAT(n, bt, wt):
    tat = [0] * n

    for i in range(n):
        tat[i] = bt[i] + wt[i]

    return tat

def find_avg_time(process, n, bt, at):
    proc_info = sorted(
        zip(process, at, bt),
        key=lambda x: x[1]
    )

    process, at, bt = zip(*proc_info)

    wt = find_wt(process, n, bt, at)
    tat = find_TAT(n, bt, wt)
```

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```
total_wt = sum(wt)
total_tat = sum(tat)

print("Execution order:", " -> ".join(map(str, process)))

print("\nProcess\tArrival\tBurst\tWaiting\tTurnaround")

Settings i in range(n):
    print(
        f"{process[i]}\t{at[i]}\t{bt[i]}"
        f"\t{wt[i]}\t{tat[i]}"
    )

print(f"\nAverage Waiting Time = {total_wt / n:.2f}")
print(f"Average Turnaround Time = {total_tat / n:.2f}")

if __name__ == "__main__":
    process = [1, 2, 3, 4]
    arrival_time = [0, 1, 2, 3]
    burst_time = [5, 3, 8, 6]

    find_avg_time(
        process,
        len(process),
        burst_time,
        arrival_time
    )
```

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Output :

Output for the given process, arrival-time, and burst-time values:

```
sayan@ubuntu: ~
sayan@ubuntu:~$ nano fcfs.py
sayan@ubuntu:~$ python3 fcfs.py
Execution order: 1 -> 2 -> 3 -> 4

Process Arrival Burst Waiting Turnaround
1 0 5 0 5
2 1 3 4 7
3 2 8 6 14
4 3 6 13 19

Average Waiting Time = 5.75
Average Turnaround Time = 11.25
sayan@ubuntu:~$
```

Result

The FCFS scheduling program was executed. The average waiting time is 5.75 time units and the average turnaround time is 11.25 time units.

SJF CPU Scheduling

Aim

To write and execute a Python program to implement the Shortest Job First (SJF) CPU scheduling algorithm and calculate the waiting time and turnaround time for each process, along with their averages.

Program

```
GNU nano 8.7.1 sjf.py *
def find_waiting_time(process, n, bt, wt):
    wt[0] = 0

    for i in range(1, n):
        wt[i] = bt[i - 1] + wt[i - 1]

def find_turnaround_time(n, bt, wt, tat):
    for i in range(n):
        tat[i] = bt[i] + wt[i]

def find_avg_time(process, n, bt):
    wt = [0] * n
    tat = [0] * n

    # Sort processes according to burst time
    for i in range(n):
        for j in range(i + 1, n):
            if bt[i] > bt[j]:
                bt[i], bt[j] = bt[j], bt[i]
                process[i], process[j] = process[j], process[i]

    find_waiting_time(process, n, bt, wt)
    find_turnaround_time(n, bt, wt, tat)

    total_wt = sum(wt)
    total_tat = sum(tat)

    print("SJF Execution Order:")
    print(" -> ".join(map(str, process)))

    print("\nProcess\tBurst Time\tWaiting Time\tTurnaround Time")

    for i in range(n):
        print(
            f"{process[i]}\t{bt[i]}\t\t"
            f"{wt[i]}\t\t{tat[i]}"
        )

    print(f"\nAverage Waiting Time = {total_wt / n:.2f}")
    print(f"Average Turnaround Time = {total_tat / n:.2f}")

if __name__ == "__main__":
    process = [1, 2, 3, 4]
    burst_time = [5, 3, 8, 6]

    find_avg_time(
        process,
        len(process),
        burst_time
    )
```

Output Screenshot

Output for the given burst-time values:

```
sayan@ubuntu: ~  
  
sayan@ubuntu:~$ nano sjf.py  
sayan@ubuntu:~$ python3 sjf.py  
SJF Execution Order:  
2 -> 1 -> 4 -> 3  
  
Process Burst Time      Waiting Time      Turnaround Time  
2         3              0                 3  
1         5              3                 8  
4         6              8                14  
3         8             14                22  
  
Average Waiting Time = 6.25  
Average Turnaround Time = 11.75  
sayan@ubuntu:~$
```

Result

The SJF scheduling program was executed successfully. The average waiting time is 6.25 time units and the average turnaround time is 11.75 time units.